

SCIENCE-BASED SPEED TRAINING / IMPLEMENTATION SPECIFICATION

SprintLab Workout Library and Season Engine

V3

A concise evidence report for the repository, test runs, conflicts, and decisions behind V3.

VERSION
3.0
CREATED
July 30, 2026
IMPLEMENTATION REVIEWED
SprintLab 1.0.0 working tree based on c00bd88
BASE COMMIT
c00bd885e64d52e624b3cf38cc15ab3 260 915 586
SOURCE OF TRUTH
Current repository behavior and passing planner regressions

Prepared from the current SprintLab working tree. V2 is background material only.

EXECUTIVE SUMMARY

Audit conclusion

The deterministic planner and performance-time verification scripts pass in the reviewed working tree. The library contains 64 Approved recommendation-eligible records and generated 28 auditable profile runs. V3 documents the implementation faithfully and records material gaps instead of rewriting the app to match V2.

Check	Result	Evidence
Planner architecture	PASS	Planner architecture scenarios passed.
Planner pathways/regressions	PASS	All release planner scenarios passed.
Performance-time ranges	PASS	Performance-time input range verification passed.
Library approval	PASS	64 Approved eligible; 0 Approved records with current validation errors.
TypeScript	FAIL	Three profile.tsx type errors involving SprintEvent=300m and a narrower helper.
Lint	INCONCLUSIVE	expo lint produced no result in the audit window and was stopped.
Profile matrix	28 generated	Beginner, trained, advanced, reduced-readiness, competition, and in-season cases.

Repository state reviewed

Property	Value
Repository	/Users/joshuaacha/Desktop/sprintlab
App version	1.0.0
Base commit	c00bd885e64d52e624b3cf38cc15ab3 260 915 586
Base commit date	July 26, 2026
Review date	July 30, 2026
Working tree	Dirty; reviewed current filesystem state, not only the base commit.
Generated evidence	docs/v3/build/v3-audit-data.json and test-logs/

Files and modules inspected

- types/domain.ts — AthleteProfile, events, season calendar, performance/profile types
- types/workout-library.ts — exact library schema and approval/filter types
- types/index.ts — PlannedWorkout, execution tracking, active/completed session structures
- data/workout-library.ts — 70 seeded workout records and prescriptions
- data/workout-sources.ts — saved research-source catalog
- utils/plan-selector.ts — day blocking, hard gates, ranking, conversion, strength pairing, planner branches
- utils/weekly-architecture.ts — phase-specific high/low role architecture
- utils/season-engine.ts — season derivation and meet-window eligibility
- utils/workout-library.ts — persistence, migration, approval validation, archive/restore
- utils/readiness.ts — green/yellow/red readiness and maximal-sprint restriction
- utils/workout-session.ts — execution-item initialization and state derivation
- utils/domain-adapters.ts — completed-session/domain-log conversion
- app/workout.tsx — active execution, per-rep/set completion, quick exercise completion

- scripts/verify-planner-architecture.ts — architecture and strength-template regressions
- scripts/verify-plan-pathways.ts — pathway, tier, schedule-load, coach/log, and History regressions
- scripts/verify-performance-time.ts — performance-time range regressions
- SprintLab_Workout_Library_and_Season_Engine_V2.md — background only
- SprintLab_40_Yard_and_General_Speed_Research.md — scientific rationale and S-source catalog

Tests run and results

Command / evidence	Result	Notes
npm run verify:planner	PASS	Architecture and all release pathway scenarios passed.
npm run verify:performance-time	PASS	Event-specific input range verification passed.
docs/v3/build/extract-v3-audit.ts	PASS	Generated 28 real selector runs and serialized all 70 library records.
npx tsc --noEmit	FAIL	Three pre-existing profile.tsx errors remain after excluding audit-script errors.
npm run lint	INCONCLUSIVE	No output/termination after extended wait; manually stopped.

Implementation/documentation conflicts

Topic	Documentation/requirement	Current code	Decision
Sprint set structure	Requested/V2-style contract expects 2×5 to remain two sets.	Current plan conversion flattens to 10 track reps.	Documented as unresolved; app not changed.
Per-side tracking	Requested contract expects per-side structure.	No typed side field or side result model.	Documented as unresolved.
Readiness tier regression	Requested rule applies across pathways.	Only Football/General branch lowers a tier.	Verified with reduced-readiness matrix; track cases fail.
Advanced track differentiation	Advanced should express a distinct higher tier.	Advanced 200/400 selects same IDs as trained; short path mixes GEN record.	Marked <u>Questionable</u> .
200m architecture	200/400 intended as one long-sprint pathway.	isLongSprint(event) is true only for 400m.	Documented as an implementation limitation.
Approval validation	Requested semantic rules reject vague or unresolved items.	Current validator is primarily structural/volume-based.	V3 separates implemented checks from planned rules.
Equipment resolution	Ideal behavior resolves equipment-specific alternatives.	Ordinary access is assumed; several item names contain unresolved “or” options.	Documented; no false claim of deterministic exercise resolution.
Source IDs	Scientific sources should be externally verifiable.	Two records use “User helper note.”	Identified as an internal authoring note, not a publication.

Profile-run decisions

Ratings across 28 runs: 21 Pass, 5 Questionable, 2 Fail. Ratings judge structural behavior, not one exact exercise choice.

Profile	Selected IDs	Result	Reason
Beginner 60m/100m — 3 days	GEN-ACC-01+STR-01, TEM-01, MAX-01+STR-02	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Beginner 60m/100m — 5 days	GEN-ACC-01+STR-01, TEM-01, MAX-01+STR-02, TEM-03, PLY-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.

Profile	Selected IDs	Result	Reason
Trained 60m/100m — 3 days	ACC-02+STR-01, TEM-01, MAX-02+STR-02	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Trained 60m/100m — 5 days	ACC-02+STR-01, TEM-01, MAX-02+STR-02, TEM-03, SED-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Advanced 60m/100m — 5 days	GEN-ACC-03+STR-01, TEM-01, MAX-02+STR-02, TEM-03, SED-04	QUESTIONABLE	Structure is coherent, but it mixes a general-speed acceleration record with shared track records.
Reduced-readiness advanced 60m/100m — 5 days	GEN-ACC-03+STR-01, TEM-01, MAX-02+STR-02, TEM-03, SED-04	FAIL	Track pathway returned the same advanced week; weekly tier regression is not applied.
Beginner 200m/400m — 3 days	ACC-01+STR-01, TEM-01, MAX-01+STR-02	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Beginner 200m/400m — 5 days	ACC-01+STR-01, TEM-01, MAX-01+STR-02, TEM-03, PLY-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Trained 200m/400m — 3 days	ACC-04+STR-01, TEM-02, MAX-02+STR-02	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Trained 200m/400m — 5 days	ACC-04+STR-01, TEM-02, MAX-02+STR-02, TEM-03, SED-04	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Advanced 200m/400m — 5 days	ACC-04+STR-01, TEM-02, MAX-02+STR-02, TEM-03, SED-04	QUESTIONABLE	Structure is coherent, but selected IDs are identical to trained and no distinct advanced long-sprint tier is expressed.
Reduced-readiness advanced 200m/400m — 5 days	ACC-04+STR-01, TEM-02, MAX-02+STR-02, TEM-03, SED-04	FAIL	Track pathway returned the same advanced week; weekly tier regression is not applied.
Beginner Football/40-Yard — 3 days	F40-ACC-01+STR-01, GEN-LOW-01, F40-MAX-01+STR-02	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Beginner Football/40-Yard — 5 days	F40-ACC-01+STR-01, GEN-LOW-01, F40-MAX-01+STR-02, CORE-02, PLY-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Trained Football/40-Yard — 3 days	F40-ACC-02+STR-01, GEN-LOW-01, F40-MAX-02+STR-02	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Trained Football/40-Yard — 5 days	F40-ACC-02+STR-01, GEN-LOW-01, F40-MAX-02+STR-02, CORE-02, F40-TRANSFER-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Advanced Football/40-Yard — 5 days	F40-ACC-03+STR-01, GEN-LOW-01, F40-MAX-03+STR-02, CORE-02, F40-TRANSFER-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Reduced-readiness advanced Football/40-Yard — 5 days	F40-ACC-02+STR-01, GEN-LOW-01, F40-MAX-02+STR-02, CORE-02, F40-TRANSFER-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Beginner General Speed — 3 days	GEN-ACC-01+STR-01, GEN-LOW-01, GEN-MAX-01+STR-02	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Beginner General Speed — 5 days	GEN-ACC-01+STR-01, GEN-LOW-01, GEN-MAX-01+STR-02, CORE-02, PLY-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Trained General Speed — 3 days	GEN-ACC-02+STR-01, GEN-LOW-01, GEN-MAX-02+STR-02	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Trained General Speed — 5 days	GEN-ACC-02+STR-01, GEN-LOW-01, GEN-MAX-02+STR-02, CORE-02, GEN-INTEGRATE-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Advanced General Speed — 5 days	GEN-ACC-03+STR-01, GEN-LOW-01, GEN-MAX-03+STR-02, CORE-02, GEN-INTEGRATE-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.

Profile	Selected IDs	Result	Reason
Reduced-readiness advanced General Speed — 5 days	GEN-ACC-02+STR-01, GEN-LOW-01, GEN-MAX-02+STR-02, CORE-02, GEN-INTEGRATE-01	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
Competition 60m/100m — 3 days	GEN-COURT-ACC-01, TEM-03, MAX-06	QUESTIONABLE	Three-role competition week is complete, but a court/limited-space acceleration record is selected for a track profile.
Competition 200m/400m — 3 days	STA-01, TEM-03, MAX-06	PASS	The generated week contains the expected pathway qualities, support structure, and explained open days.
In-season Football with Tue/Thu practices and Sat game	GEN-MICRO-01, GEN-LOW-01, CORE-01	QUESTIONABLE	Practice/game blocking works, but the remaining three sessions contain no meaningful loaded-strength exposure.
In-season General Speed with Tue/Thu practices and Sat game	GEN-MICRO-01, GEN-LOW-01, CORE-01	QUESTIONABLE	Practice/game blocking works, but the remaining three sessions contain no meaningful loaded-strength exposure.

Decisions made

- Used the current repository and test output as implementation authority; V2 was not copied forward.
- Did not modify planner or application behavior to make it agree with the requested ideal.
- Documented broader AthleteProfile fields only when current selection code reads them.
- Separated daily readiness launch guidance from weekly profile-level tier regression.
- Listed all 64 Approved/selectable records with current prescriptions and all six nonselectable records separately.
- Used saved R and S sources for rationale without implying endorsement.
- Marked actual defects and incomplete semantics as limitations even when planner scripts passed.

Could not be verified

- A clean full TypeScript build, because three profile.tsx type errors remain.
- A completed lint run, because the configured lint process did not return a result.
- Physical-device rendering and interaction for every workout prescription; this audit is implementation/test based.
- Scientific support for the internal “User helper note” source label.
- Set-preserving sprint execution or per-side logging, because those structures are not implemented.

FINAL AUDIT JUDGMENT

V3 is a faithful implementation specification. It is not a claim that every requested planner/execution behavior is complete.